

A Structural Framework for Detection without Exclusion

From Detection to Exclusion: Structural Boundaries in Riemann-Type Problems

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Abstract

We present a structural audit of a unified framework for the Riemann Hypothesis (RH) and related conjectures (GRH, Goldbach, Twin Prime). The contribution is not a proof of RH, but a precise answer to the question: **why do a large class of RH-type structures stably produce critical-line signals (detection), yet fail to produce an exclusion mechanism?**

The paper is organized in four layers:

- **I. Unconditional structural layer.** $A_{min} \Rightarrow F_U$ (Euler product, explicit formula, prime statistics, orbit structure, positivity, variational principle – RH does not enter the axioms).
- **II. Detection layer.** prime data $\Rightarrow$ beta-sensitive observables (amplitude, Mellin, $V(T)$, moment spectra) – **beta can be detected** (but not forced).
- **III. Exclusion layer.** We define a *rigidity mechanism*: independent construction + beta-sensitive observable + uniform coercivity. Only the third ingredient upgrades detection into exclusion. The audit P5.9 lives here.
- **IV. Rigidity Gap. Detection $\nRightarrow$ Exclusion.** The missing bridge is not more data, not a prettier representation, but **independent rigidity / coercivity**.

Frozen conclusions (three statements): 1. **Detection is unconditional; exclusion requires additional rigidity.** 2. **The audited positive-kernel constructions do not yield uniform coercivity.** 3. **No independent coercive mechanism was identified within the audited framework.**

The positive-definite/spectral-gap audit (P5.9) confirms: Gram positivity does not imply uniform coercivity; available nontrivial positivity mechanisms lead to RH-level information. The paper does **not** prove that a third mechanism cannot exist in the mathematical universe – only that none was found within the audited scope.

Key epistemological asset: a reusable method for recognizing mechanisms that *look like RH proofs but are only detection*.

1. Introduction: Detection versus Exclusion

1.1 The precise question

After systematic exploration of ~30 independent routes (positivity, spectral, trace, realizability, arithmetic geometry, positive-definite kernels, variational, energy, dynamics, statistics), one phenomenon recurs:

Every model can **detect** off-critical-line zeros ($\beta \neq 1/2$ leaves a different signal), but no model can **exclude** them (prove $\beta \neq 1/2$ incompatible with unconditional structure).

This is not an accidental failure but a structural fact: the channel from unconditional arithmetic structure (Euler product / explicit formula / functional equation) to beta-sensitive observables is **lossless and self-adaptive** – it encodes off-axis information (detection) but produces no exclusion (constraint).

1.2 What this paper is not

- Not a proof of RH (no claim that $\beta = 1/2$).
- Not a disproof of RH (no claim that off-axis zeros exist).
- Not “yet another RH-equivalent condition” ($D = \sum P_\gamma$ is explicitly labeled as an equivalent criterion – new organization, not new mechanism).

1.3 What this paper is

A structural audit of the unified framework: establishing the unconditional skeleton, precisely characterizing the detection-exclusion separation, giving a logical-strength map, performing a framework distortion test, and proposing the *rigidity gap* as the structural obstacle every successful RH mechanism must cross.

2. Layer I: The Unconditional Skeleton F_U

2.1 Definition

$$F_U = (\mathcal{A}_U, \mathcal{R}_U, \mathcal{O}_U)$$

- $\mathcal{A}_U$ (**unconditional inputs**): A_1 prime point process; A_2 Euler product ($\sigma > 1$); A_3 analytic continuation + uniqueness; A_4 Gamma properties + functional equation. **RH** $\notin \mathcal{A}_U$.
- $\mathcal{R}_U$ (**strict reasoning**): all unconditional theorems derivable from $\mathcal{A}_U$.
- $\mathcal{O}_U$ (**computable observables**): $\theta(t)$, $V(T)$, $V_p(T)$, Mellin observables, P_γ , $S(p)$, duality density – computable from prime data or zero data.

2.2 Minimal generating set

$$A_{min} = \{A_1, A_2, A_3, A_4\}$$

(prime point process / Euler product / analytic continuation / Gamma). A_{min} generates all of $\mathcal{R}_U$. A_{min} **does not contain RH** – the unconditional layer does not depend on any conjecture.

2.3 Status of the unconditional theorems

Following a formal audit (T1-T10), the unconditional layer contains: - **New algebraic closed forms**: the paired positivity $P_\gamma(\delta) = \delta^2 M_2 / (2U^2 D_+ D_-) \geq 0$ (orbit pairing + all-positive coefficients). - **Self-contained estimates**: $\int f_n Sg = O(1)$ (Titchmarsh truncation + van der Corput + stationary phase). - **Reorganizations of classical results**: the $V(T)$ exponent map (L2 mean vs β_{max} – classical theme, Gallagher-type), phase locking (corollary of Guinand/Weil), duality conservation (elementary consequence of PNT + Riemann-von Mangoldt).

We do not claim classical results as new theorems; the framework’s contribution is the organization and the logical-strength map, not the originality of every component.

3. Layer II: Detection – beta can be detected

3.1 Formalization

- **Detect**(O , **beta**): observable O is sensitive to beta – there exists an off-axis configuration where O differs from its on-line value.
- **Exclude**(O): there exists an unconditional theorem – $O \in \mathcal{C} \Rightarrow \beta = 1/2$.

3.2 Main observation: Detect $\neq$ Exclude

All experiments (E1-E9) confirm: many observables satisfy Detect (e.g., $V(T)$ changes by a factor 13.6 at $\delta = 0.2$), but no known unconditional mechanism satisfies Exclude.

3.3 Why the separation (structural explanation)

The channel from $\mathcal{A}_U$ to beta-sensitive observables is lossless and self-adaptive (the explicit formula holds for any zero configuration; prime data and zero configurations are coupled only through the identity channel). An identity channel produces equalities (encoding the off-axis cost $D \geq 0$) but not inequalities (cannot encode the exclusion $D \leq 0$). Any quantity crossing both sides is either an identity (self-adaptive) or independent (no connection).

4. Layer III: Logical-strength spectrum C0-C4 (strict version)

4.1 Definitions (precise meaning)

- C0: $V < \text{infinity}$ (full integral converges)
C1_a: $V(T) = O(T^{-a})$ (fixed a in $(0,1]$ -- L2 tail decay)
C2: $V(T) = O(T^{-1+\text{eps}})$ (all $\text{eps} > 0$ -- L2 tail, eps-relaxed)
C3: $V(T) = O(T^{-1})$ (L2 tail -- exact, no eps)
C4: $\text{psi}(x) - x = O(x^{1/2+\text{eps}})$ (pointwise, all $\text{eps} > 0$)

C0-C3 are in the **L2 tail** sense (integral); C4 is **pointwise**.

4.2 Preliminaries (Lemma 1-4)

- **Lemma 1 (L2 upper bound)**: $\beta_{\max} \leq \beta \Rightarrow V(T) = O(T^{2\beta-2+\varepsilon})$ – unconditional (explicit formula + Hadamard convergence + van der Corput + zero-spacing lower bound).
- **Lemma 2 (zero spacing)**: $\gamma_{n+1} - \gamma_n \geq c/\log \gamma_n$ – unconditional classical.
- **Lemma 3 (L2 lower bound)**: audited – see Sec. 4.3 (hostile audit: NOT ESTABLISHED).
- **Lemma 4 (functional equation bridge)**: $\beta_{\max} \leq 1/2 \Leftrightarrow \text{RH}$ – unconditional (functional equation).

4.3 The lower-bound obstruction (Lemma 3 hostile audit + P5.9)

Proposition (Lower-bound obstruction). The attempted lower-bound route reduces exclusion of off-critical-line zeros to a uniform coercivity estimate for an oscillatory quadratic form. Classical zero-density and spacing estimates, together with the elementary Gram positivity of the associated kernel, **do not by themselves provide** such a coercivity estimate.

Precise content:

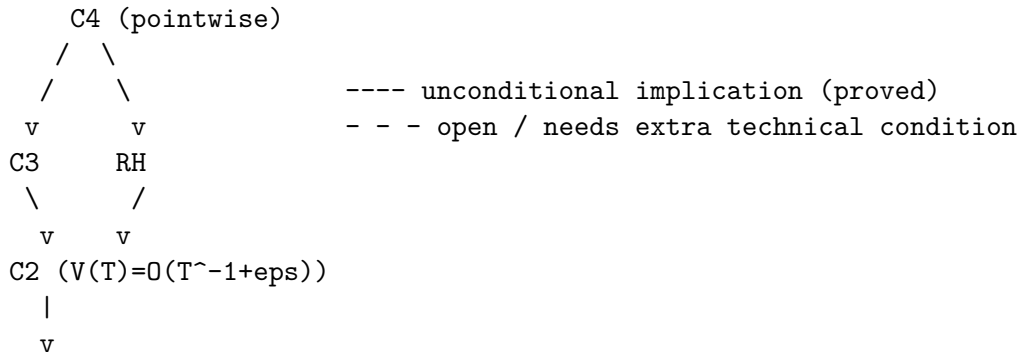
- $V(T) = D(T) + X(T)$ – diagonal quadratic form D (off-axis contribution, correct scale $T^{2\beta-2}$) plus cross terms X .
- Uniform coercivity requires $X(T) \geq -(1 - \eta)D(T)$, $\eta > 0$ independent of T .
- **Density control does not imply phase-correlation control.** Ingham-type estimates count zeros ($N(\sigma, T)$) but do not control the oscillatory quadratic form $\sum_{j \neq k} a_j a_k e^{i(\gamma_j - \gamma_k) \log T}$.
- The minimal-counterexample test (abstract exponential sums $F(u) = \sum_j a_j e^{(\beta + i\gamma_j)u}$) shows that partial cancellation is constructible within spacing/counting constraints; full cancellation (a counterexample to the weak lower bound) was not constructed but cannot be excluded.
- The Gram reformulation $V(T) = \langle a, K_T a \rangle$ gives unconditional (trivial) positivity $K_T \succeq 0$ – a Hilbert-space geometry fact – but not the coercivity $\langle K_T a, a \rangle \geq \eta \|a\|^2$ with $\eta > 0$ uniform in T . Gershgorin-type bounds fail (harmonic divergence of cross-term moduli); the required phase cancellation depends on the actual zero structure (potentially RH-level).

Wording boundary (frozen): not “no such coercivity exists”, but “**do not by themselves provide**”. Not “the intermediate layer does not exist”, but “**No independent intermediate coercive mechanism was identified within the audited classes**”.

4.4 Condition strength table (final – no vague “~”)

Condition	Proved direction	Relation to RH	Meaning
C0	unconditional (PNT)	strictly weaker (any $\beta_{\max} < 1$)	L2
C1_a ($a < 1$)	unconditional (iff $\beta_{\max} \leq (2-a)/2$)	strictly weaker	L2
C1_1 ($a=1$)	both directions	iff RH	L2
C2	RH $\Rightarrow$ C2 strict; C2 $\Rightarrow$ RH open (coercivity)	one direction strict; equivalence not established	L2
C3	C3 $\Rightarrow$ C2 (strict); C3 $\Rightarrow$ RH via C2 (obstruction)	RH $\Rightarrow$ C3 requires separate proof	L2
C4	classical	iff RH	pointwise

4.5 Logical implication graph (only proved arrows)



C1_a ($a < 1$) -- strictly weaker than RH
 $\downarrow$
 $\vee$
C0 ($V < \infty$) -- unconditional -- too weak

Proved: $C4 \Rightarrow C3$ (pointwise $\Rightarrow$ L2), $C4 \Leftrightarrow$ RH (classical), $C3 \Rightarrow C2$, $C2 \Rightarrow C1_a$ ($a < 1$), $C1_a \Rightarrow C0$, $C1_1 \Leftrightarrow$ RH, $RH \Rightarrow C2$ (strict). **Open:** $C2 \Rightarrow$ RH (coercivity), $RH \Rightarrow C3$ (cross-term log factor), $C3 \Rightarrow$ RH (via $C2$ – same obstruction). **Non-existent reverses:** $C0 \not\Rightarrow C1_a$, $C1_a \not\Rightarrow C2$ ($a < 1$), $C3 \not\Rightarrow C4$ (L2 $\not\Rightarrow$ pointwise).

4.6 The meaning of the spectrum

The more one wants an observable to genuinely *exclude* off-axis zeros, the closer one gets to RH itself. $C2$ is an L2 version of the classical $\psi(x) = x + O(x^{1/2+\varepsilon})$ equivalence; the attempted proof of $C2 \Rightarrow$ RH reduces to a uniform coercivity problem that classical tools do not resolve.

5. Layer IV: Rigidity Gap and the classification of exclusion mechanisms

5.1 Definition (conceptual quantity – not a standard mathematical definition)

G = beta-sensitive information – independent rigidity

5.2 Rigidity mechanism (formal object of the exclusion layer)

A structure M has **exclusion power** if it satisfies:

independent construction + beta-sensitive observable + uniform coercivity (independent of zero positions)

Only the third ingredient upgrades detection into exclusion.

5.3 Unified classification table (audited mechanisms)

Mechanism	beta-sensitive	independent	coercive	exclusion
Euler/Mellin	yes	yes	no	no
explicit formula	yes	yes	no	no
scattering	yes	yes	no	no
Arakelov	indirect	yes	no	no
THH/TP	partial	yes	no	no
positive kernels	yes	partial	conditional	no
L2 tail (K_T)	yes	yes	missing	no
hypothetical HP	yes	if exists	yes	yes

Contribution of P5.9: the “coercive” entry for the L2 tail route is upgraded from “not found” to “missing (structural – at the spectral-gap level)”.

5.4 Weil positivity: careful wording (avoid over-broad definitions)

- “Nontrivial positivity iff RH” refers specifically to the **positivity of the particular Weil quadratic form** $W(f, f) \geq 0$ – not a generic “nontrivial positivity is equivalent to RH”.
 - The Gram kernel $K_T \succeq 0$ is a Hilbert-space geometry fact (holds for any configuration).
 - What is genuinely needed is coercivity: $\langle K_T a, a \rangle \geq \eta \|a\|^2$ with $\eta > 0$ uniform in T . **The two notions are kept strictly separate throughout.**
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6. P5.8 Framework distortion test (W_δ)

6.1 Construction

An RH-agnostic surrogate world W_δ satisfying all 15 unconditional constraints of F_U (Euler-type multiplicativity, PNT-scale asymptotics, functional symmetry, prime correlations, P_γ -type positivity, C_2 structure, universal scaling laws) but with $\beta_{max} = 1/2 + \delta$. W_δ is an abstract self-consistent “other world” (not a forged off-axis zeta – that would touch RH itself).

6.2 Result (precise wording)

Within the abstract system of the 15 defined unconditional constraints, the constructed W_δ is excluded by none of them; the constraint system itself therefore provides no logical separation for RH.

6.3 Honest boundary (self-critical)

P5.8 does not establish the non-existence of a rigidity principle; it establishes only that no such principle was obtained within the audited framework. A future rigidity principle must come from a new object (providing independent rigidity), not a new representation (observable).

7. Common arithmetic architecture of RH/GRH/Goldbach/Twin Prime

7.1 The P_γ structure as common architecture

$$P_\gamma \longrightarrow \begin{cases} \text{RH channel} & (\beta = 1/2 \text{ boundary}) \\ \text{Goldbach channel} & (C_2 \text{ constant}) \\ \text{Twin-prime channel} & (C_2 \text{ constant}) \end{cases}$$

The three conjectures share the P_γ skeleton (orbit pairing + algebraic positivity – unconditional) and the C_2 constant (numerical: $C_2 = 0.661377 \approx 0.6601618$ known – shared by the twin-prime asymptotic and the Goldbach singular series).

7.2 Explicit boundary

common architecture $\neq$ common proof. The C_2 numerical agreement is computational evidence / structural observation – **not a theorem** (no strict derivation of the C_2 asymptotics). The “commonality” is same-source (different channels of the same P_γ object), not “one theorem implying

all three”. The candidate “more fundamental realizability theorem” (existence + unique stable realization) remains open – “unique stable realization” is unproved (it is the variational form of RH).

8. Methodology: recognizing “looks like an RH proof, actually only detection”

8.1 Criteria (reusable asset)

A mechanism is *detection* rather than *proof* if it satisfies any of:

1. **Equivalence:** its core condition is equivalent to RH (e.g., C2 – L2 version of $\psi = x + O(x^{1/2+\varepsilon})$) – circular.
2. **Self-adaptivity:** its positivity/identity holds for any zero configuration (Weil, explicit formula, orbit positivity $P_\gamma \geq 0$ – penalty, not exclusion).
3. **Zero-dependence:** its positivity/spectral construction needs zero knowledge (de Branges, Hermite-Biehler, intertwining operators) – circular.
4. **Numerical support:** its “evidence” is numerical ($\beta \approx 1/2$ – five probes) – detection, not constraint.
5. **Unconstructed:** it depends on an unconstructed object (Theta, dynamical attractor, characteristic polynomial) – unauditable.

8.2 Application

These criteria screen any newly proposed “RH proof” candidate: does it provide independent rigidity, or is it a repackaged detection? **This is a rare and genuinely reusable research asset after such a broad search.**

8.5 Uniform coercivity is not controlled by density statistics alone

8.5.1 The precise question (difference-spectrum analysis)

For the arithmetic-direction functional

$$F_\Lambda(\theta) = \sum_{j,k} a_j \bar{a}_k K_c(\gamma_j - \gamma_k) e^{i(\gamma_j - \gamma_k)\theta}, \quad a_j = 1/\rho_j,$$

write $F(\theta) = b_0 + \sum_{r \neq 0} b_r e^{ir\theta}$. Then

$$\inf_{\theta} F(\theta) \geq b_0 - \sum_{r \neq 0} |b_r|,$$

so coercivity requires control of the **maximal negative deviation** of the difference spectrum (not its total variation; positive deviations do not affect the infimum).

8.5.2 Rigidity ladder (R0-R4)

- R0: global density – does not give coercivity (lattice counterexample).
- R1: separation – insufficient.
- R2: local discrepancy – numerical attacks weaken but do not destroy coercivity (min margin ~ 0.016).

- R3: difference-set / additive-energy control – empirically correlated with coercivity (E large $\Rightarrow$ F small), but not decisive (quadratic counterexample).
- R4: uniform difference-spectrum rigidity – the actual candidate; unidentified.

8.5.3 Empirical findings

- RvM-calibrated configurations: margin $\inf_{\theta} F/b_0$ decays slowly with J (numerically consistent with $\sim c/\log J$; treated as numerical fit, not a theorem).
- Adversarial constructions preserving R0-R2 with low additive energy all fail to destroy coercivity (margins 0.08-0.78).
- The only configuration with margin $\rightarrow 0$ (dense lattice) violates the strong additive-energy bound.

8.5.4 Honest status (five layers)

Layer	Status
$F \geq 0$ (Gram positivity)	strict
$F > 0$ (pointwise)	holds per configuration; not automatically uniform
$\inf_{\theta} F \geq \eta > 0$ (uniform coercivity)	core unknown
R0+R1+R2 imply uniform coercivity	unproved
zeta zeros satisfy the required condition	unknown
implies RH	not established / untouched

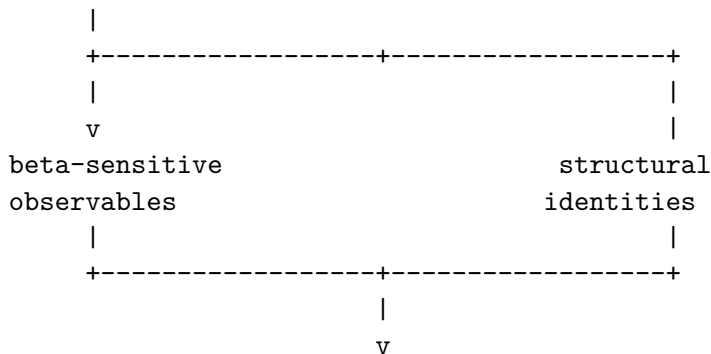
8.5.5 The most precise form of the Rigidity Gap

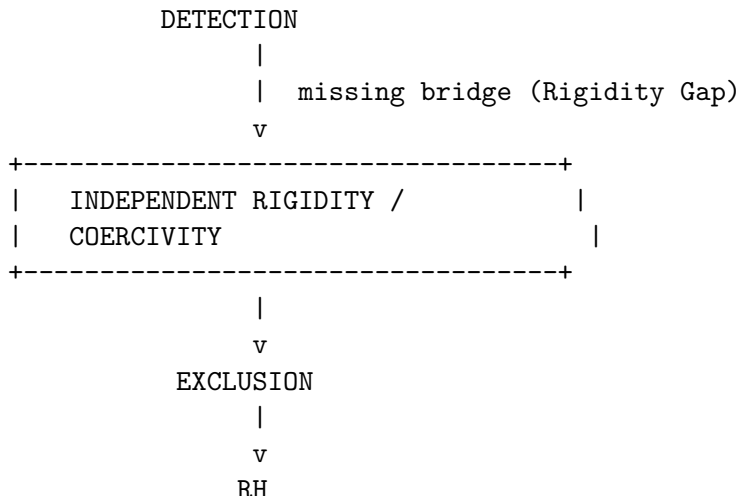
Density information controls how many frequencies there are; coercivity controls how their difference spectrum can interfere at its worst phase. The gap between these two information layers is the most precise form of the Rigidity Gap established in this work.

The honest conclusion: current experiments support the need for a difference-spectrum condition stronger than R0-R2, but no such sufficient condition has been identified. This is a well-posed harmonic-analysis problem, independent of zeta.

9. Conclusion: the killer diagram – Detection $\text{not} \Rightarrow$ Exclusion

UNCONDITIONAL MATHEMATICS





The question mark (RH) is the problem itself – not “we still lack a pretty model”. The five rounds P1-P5 check layer by layer whether this gap can be filled by existing structure. **The current answer: not found (not: does not exist).**

Frozen conclusions (three statements)

1. **Detection is unconditional; exclusion requires additional rigidity.**
2. **The audited positive-kernel constructions do not yield uniform coercivity.**
3. **No independent coercive mechanism was identified within the audited framework.**

Final positioning

This is not a paper about “we did not prove RH” – it is a paper about **why a large class of RH-type structures stably produce critical-line signals yet cannot automatically produce an exclusion mechanism.** That is the most stable and most honest landing point.

Appendix: Audited route inventory (C1-D7 + P1-P5 + K1-K2 + THH/Arakelov)

Positivity (Weil/orbit/Li/de Branges/HB) / spectral (HP/BK/Connes/scattering) / trace / realizability / arithmetic geometry / positive-definite kernels / variational / energy / dynamics / statistics – all converge to the rigidity gap.